

1. Business Problem Document

Project Title

Sales Insight: Customer Shopping Behavior Analysis

Background

A retail company operates across multiple product categories and sales channels (online and offline). Recently, management has observed shifts in customer purchasing behavior influenced by demographics, discounts, seasons, reviews, and subscription models. To stay competitive, the company needs data-driven insights to understand these changes and optimize its sales and marketing strategies.

Business Problem Statement

The company lacks a consolidated, analytical view of customer shopping behavior. Decisions related to promotions, product positioning, and customer engagement are currently made with limited insight into what truly drives purchases and repeat buying.

Core Business Question:

How can the company leverage sales and customer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?

Objectives

- Analyze customer demographics and purchasing patterns
- Identify key revenue drivers (discounts, subscriptions, shipping type, reviews)
- Segment customers based on loyalty and purchase behavior
- Compare performance across product categories and seasons
- Enable stakeholders to make informed, data-driven decisions

Scope of the Project

- Dataset of ~3,900 customer purchase transactions
- Analysis across demographics, products, pricing, and promotions
- Use of Python for data preparation, SQL for analysis, and Power BI for visualization

Key Stakeholders

- Sales Team
- Marketing Team
- Product Managers
- Executive Management

Deliverables

1. Cleaned and modeled dataset (Python)

2. Business-focused SQL analysis
3. Interactive Power BI dashboard
4. Analytical report with insights and recommendations
5. Project repository with all artifacts

Expected Business Value

- Improved targeting of promotions and discounts
- Increased customer retention through loyalty and subscription insights
- Better product placement and inventory planning
- Higher revenue through optimized marketing strategies